

PHITSANULOK, Thailand

WMO: 483780

Lat: 16.7961N

Lon: 100.2758E

Elev: 150

StdP: 14.62

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.7	64.5	51.1	56.0	72.7	54.3	63.2	74.4	11.2	81.6	9.9	82.6	1.2	10	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.9	100.4	78.1	98.6	78.1	96.8	77.9	81.7	91.6	81.0	90.7	80.7	90.2	5.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	153.6	85.2	78.9	151.0	84.8	78.3	148.0	84.5	45.6	92.2	45.0	90.9	44.4	90.5	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.3	9.7	8.4	56.8	103.1	3.6	2.2	54.2	104.7	52.1	106.0	50.0	107.2	47.4	108.8	
(5)				53.7	84.0	3.4	1.3	51.3	84.9	49.3	85.7	47.4	86.4	44.9	87.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	77.8	81.3	85.4	88.4	86.9	85.4	84.2	83.6	83.6	83.4	81.2	77.7
	DBStd	4.33	3.64	3.51	3.44	3.13	3.28	2.21	2.15	1.89	1.74	2.10	2.95	4.15
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	0	0	0	0	0	0	0	0	0	0	1
	CDD50	12129	860	876	1096	1153	1144	1063	1059	1041	1009	1034	937	857
	CDD65	6656	396	456	631	703	679	613	594	576	559	569	487	393
	CDH74	76514	3540	4798	7777	9684	8870	7542	6827	6448	6217	6306	4998	3507
	CDH80	35181	1414	2263	4076	5602	4687	3515	2844	2511	2387	2496	2046	1340
Wind	WSAvg	2.9	2.4	2.7	3.3	3.7	3.5	3.5	3.1	3.1	2.6	2.5	2.4	2.4
Precipitation	PrecAvg	53.1	0.3	0.5	1.2	2.1	7.2	6.9	7.5	9.6	10.1	5.8	1.4	0.4
	PrecMax	73.0	2.8	3.5	6.5	7.7	21.3	13.9	18.3	17.4	18.8	14.6	7.4	4.2
	PrecMin	36.6	0.0	0.0	0.0	0.0	0.4	0.0	2.6	3.9	2.6	0.2	0.0	0.0
	PrecStd	9.5	0.6	0.7	1.4	1.8	3.9	3.2	3.0	3.5	3.9	3.4	1.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.8	96.6	100.2	103.7	102.6	98.5	96.1	93.7	93.5	93.9	94.6	93.1
		MCWB	74.0	74.5	75.5	76.9	78.8	78.2	79.2	79.6	79.6	79.0	77.6	75.7
	2%	DB	91.1	94.6	98.2	100.9	99.8	95.6	93.5	92.2	91.8	92.5	92.7	91.1
		MCWB	73.6	75.1	76.1	78.2	79.1	79.2	79.4	79.4	79.4	78.6	77.0	74.8
	5%	DB	89.5	93.0	96.5	99.2	97.2	93.6	91.8	91.0	90.9	91.3	91.2	89.4
		MCWB	72.8	75.0	76.4	78.3	79.1	79.4	79.2	79.2	79.3	78.3	76.3	73.9
	10%	DB	87.6	91.2	94.8	97.3	94.9	91.8	90.3	89.5	89.5	89.8	89.5	87.4
		MCWB	72.0	74.6	76.5	78.4	79.3	79.3	79.0	79.1	79.2	78.0	75.6	72.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.1	79.7	80.7	81.9	82.6	81.9	81.6	81.8	81.8	81.6	80.4	78.9
		MCDB	87.9	91.5	93.8	96.0	93.7	91.0	90.5	89.4	89.1	89.5	89.7	87.6
	2%	WB	76.5	78.3	79.6	81.0	81.3	81.0	80.7	80.9	80.9	80.6	79.2	77.2
		MCDB	85.6	89.9	92.8	94.9	92.6	90.2	89.4	88.4	88.2	88.5	87.6	85.9
	5%	WB	75.4	77.4	78.7	80.3	80.7	80.5	80.1	80.1	80.3	79.9	78.4	75.9
		MCDB	84.1	88.5	91.9	94.1	92.0	89.7	88.4	87.5	87.4	87.4	86.3	84.6
	10%	WB	74.3	76.4	77.9	79.6	80.0	79.9	79.6	79.7	79.8	79.3	77.5	74.7
		MCDB	83.0	87.0	90.7	93.2	90.9	88.7	87.6	86.9	86.7	86.3	85.1	83.4
Mean Daily Temperature Range	5% DB	MDBR	20.3	20.6	19.6	18.9	16.1	14.1	12.9	12.3	12.4	13.7	16.6	18.9
		MCDBR	22.0	22.0	21.4	20.4	18.9	16.1	14.8	14.2	14.1	15.4	18.1	19.7
		MCWBR	8.0	8.1	7.3	6.4	5.5	4.9	4.9	5.1	5.0	5.0	5.8	6.7
	5% WB	MCDBR	18.3	19.7	19.6	18.8	16.6	14.3	13.5	13.0	13.1	13.9	15.7	17.5
		MCWBR	7.2	7.9	7.2	6.5	5.7	5.1	5.1	5.3	5.1	5.1	5.6	6.4
Clear-Sky Solar Irradiance	taub	0.488	0.590	0.684	0.627	0.506	0.448	0.437	0.446	0.485	0.552	0.472	0.477	
	taud	2.063	1.827	1.649	1.814	2.183	2.367	2.396	2.358	2.234	1.998	2.201	2.115	
	Ebn at Noon	252	232	215	229	255	268	271	271	261	239	256	251	
	Edh at Noon	50	67	82	70	48	39	38	40	45	56	44	47	
All-Sky Solar Radiation	RadAvg	1489	1641	1712	1890	1766	1624	1521	1472	1482	1506	1535	1449	
	RadStd	86	69	101	89	122	109	88	76	83	90	72	55	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+196	+198

Nomenclature: See separate page